

7 Parkinson's disease occurs when dopamine-producing neurones die.

- (a) The most obvious motor symptoms of Parkinson's disease include muscle tremors and muscle rigidity.

These motor symptoms are linked to the loss of dopamine-producing neurones in the brain.

- (i) One drug used to treat Parkinson's disease is L-DOPA.

Which of the following statements explains why L-DOPA can be used to treat Parkinson's disease?

(1)

- ☐ **A** L-DOPA acts on a different post-synaptic receptor to dopamine
- ☐ **B** L-DOPA crosses the blood-brain barrier and is then converted to dopamine in the brain
- ☐ **C** L-DOPA is converted to dopamine in the blood and then crosses the blood-brain barrier
- ☐ **D** L-DOPA is identical to dopamine

(ii) Explain why the release of reduced quantities of dopamine by presynaptic neurones could result in motor symptoms.

(3)

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(b) It has been suggested that in Parkinson's disease dopamine-producing neurones die because of a failure to release dopamine.

Suggest how the release of dopamine from presynaptic neurones could be inhibited.

(2)

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(c) Describe how microarrays and bioinformatics could be used to investigate the genetic basis of Parkinson's disease.

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(Total for Question 7 = 10 marks)
